GREEDY APPROACH - NUMERICAL SOLUTION

Complete Step-by-Step Walkthrough

PROBLEM SCENARIO

Setup

- **5-armed bandit** with 20 total plays
- **True reward distributions** (unknown to agent):
  - Arm 0: Probability 0.30 → Average reward 3.0
  - Arm 1: Probability 0.50 → Average reward 5.0
  - Arm 2: Probability 0.60 → Average reward 6.0 **OPTIMAL**
  - Arm 3: Probability 0.20 → Average reward 2.0
  - Arm 4: Probability 0.45 → Average reward 4.5
- **Maximum payout per pull:** 10
- **Strategy:** Pure greedy (always select argmax Q)
- **No exploration:** Once one arm seems best, exploit it

ALGORITHM DESCRIPTION

Greedy Selection Rule

```
At each play t:
  1. Calculate Q(a) for all arms (average of past rewards)
  2. Select arm with highest Q: a* = argmax Q(a)
  3. Pull arm, observe reward R_t
  4. Update Q(a*) with new reward
```

Q-Value Update Formula

```
Q_new(a) = (Q_old(a) × N_old(a) + R_t) / N_new(a)

Where:
  Q_old(a) = previous estimate
  N_old(a) = times arm was pulled before
  R_t = reward received
  N_new(a) = N_old(a) + 1
```

COMPLETE NUMERICAL WALKTHROUGH

INITIAL STATE (Play 0)

Arm	Q(a)	N(a)	Status
0	0.0	0	Untried
1	0.0	0	Untried
2	0.0	0	Untried
3	0.0	0	Untried
4	0.0	0	Untried

Note: All Q values tied at 0.0 → Select Arm 0 arbitrarily

DETAILED PLAYS 1-20

PLAY 1

Greedy Selection: All arms tied at Q=0 → Pick Arm 0

Simulated Reward: Arm 0 has p=0.30 - Random draw: 0.42 (no payout) - **Received: 0**

Q-Value Update for Arm 0: - Old N(0) = 0 - New N(0) = 0 + 1 = 1 - Old Q(0) = 0.0 - New Q(0) = (0.0 × 0 + 0) / 1 = **0.0**

State After Play 1:

Arm	Q(a)	N(a)
0	0.0	1
1	0.0	0
2	0.0	0
3	0.0	0
4	0.0	0

Next Selection: Still tied; pick Arm 1 arbitrarily

PLAY 2

Greedy Selection: All tied at Q=0 → Pick Arm 1 (next untried)

Simulated Reward: Arm 1 has p=0.50 - Random draw: 0.38 (payout!) - **Received: 10**

Q-Value Update for Arm 1: - Old N(1) = 0 - New N(1) = 0 + 1 = 1 - Old Q(1) = 0.0 - New Q(1) = (0.0 × 0 + 10) / 1 = **10.0**

State After Play 2:

Arm	Q(a)	N(a)
0	0.0	1
1	10.0	1
2	0.0	0
3	0.0	0
4	0.0	0

Next Selection: Arm 1 has highest Q = 10.0 → **LOCK IN**

PLAY 3

Greedy Selection: argmax Q = Arm 1 (Q=10.0)

Simulated Reward: Arm 1 has p=0.50 - Random draw: 0.45 (payout!) - **Received: 10**

Q-Value Update for Arm 1: - Old N(1) = 1 - New N(1) = 1 + 1 = 2 - Old Q(1) = 10.0 - New Q(1) = (10.0 × 1 + 10) / 2 = 20/2 = **10.0**

State After Play 3:

Arm	Q(a)	N(a)
0	0.0	1
1	10.0	2
2	0.0	0
3	0.0	0
4	0.0	0

Next Selection: Arm 1 still best → Continue

PLAY 4

Greedy Selection: Arm 1 (Q=10.0 highest)

Simulated Reward: Arm 1 has p=0.50 - Random draw: 0.68 (no payout) - **Received: 0**

Q-Value Update for Arm 1: - Old N(1) = 2 - New N(1) = 2 + 1 = 3 - Old Q(1) = 10.0 - New Q(1) = (10.0 × 2 + 0) / 3 = 20/3 = **6.67**

State After Play 4:

Arm	Q(a)	N(a)
0	0.0	1
1	6.67	3
2	0.0	0
3	0.0	0
4	0.0	0

Next Selection: Still Arm 1 (6.67 > 0.0)

PLAYS 5-20 SUMMARY

Key Observation: Arm 1 remains highest Q value throughout. Greedy algorithm locks in and never explores other arms!

Play	Arm	Reward	Q(1) Update	New Q(1)	N(1)	Total Reward
5	1	10	(6.67×3+10)/4	8.33	4	30
6	1	0	(8.33×4+0)/5	6.67	5	30
7	1	10	(6.67×5+10)/6	7.78	6	40
8	1	10	(7.78×6+10)/7	8.18	7	50
9	1	0	(8.18×7+0)/8	7.16	8	50
10	1	10	(7.16×8+10)/9	7.85	9	60
11	1	0	(7.85×9+0)/10	7.07	10	60
12	1	10	(7.07×10+10)/11	7.35	11	70
13	1	10	(7.35×11+10)/12	7.65	12	80
14	1	0	(7.65×12+0)/13	7.07	13	80
15	1	10	(7.07×13+10)/14	7.43	14	90
16	1	10	(7.43×14+10)/15	7.73	15	100
17	1	0	(7.73×15+0)/16	7.24	16	100
18	1	10	(7.24×16+10)/17	7.47	17	110
19	1	10	(7.47×17+10)/18	7.70	18	120
20	1	0	(7.70×18+0)/19	7.30	19	120

FINAL STATE (After Play 20)

Q-Value Estimates

Arm	Final Q	N(a)	Arms Tried	Total Reward from Arm
0	0.0	1	Once	0
1	7.30	19	19x	~**95**
2	0.0	0	Never	0
3	0.0	0	Never	0
4	0.0	0	Never	0

Performance Metrics

Metric	Value
Total Reward	95
Average Reward per Play	95/20 = **4.75**
Arms Explored	2 out of 5 (40%)
Arms Never Tried	3 out of 5 (60%)
Best Arm Found	**NO**
Plays on Optimal (Arm 2)	0
Plays on Greedy Choice (Arm 1)	19

ANALYSIS: WHY GREEDY FAILED

The Problem

Greedy got unlucky early: 1. First try: Arm 0 → reward 0 (disappointing) 2. Second try: Arm 1 → reward 10 (lucky!) 3. **Result:** Q(1) = 10 > all others → locked in

Never discovered the true best arm (Arm 2): - Arm 2's true mean = 6.0 - But greedy never tried it - Final Q(2) = 0.0 (never pulled)

Opportunity Cost

If greedy had found and exploited Arm 2 (mean 6.0):

```
Optimal: Pull Arm 2 twenty times
Expected total: 6.0 × 20 = 120

Actual with Arm 1:
Expected total: 5.0 × 20 = 100

Difference (regret): 120 - 95 = 25 points lost!
```

The Core Issue

Greedy has no exploration mechanism. Once an arm appears good through lucky early pulls, it dominates forever. This is called **local optimum** or **getting stuck**.

KEY INSIGHTS

What Greedy Teaches Us

1. **Exploration is essential**
    - Without trying other options, agent gets stuck
    - One lucky pull locks in decision forever
2. **Early luck matters**
    - Arm 1 got lucky (payout on plays 2-3)
    - Arm 2 might have gotten unlucky if tried early
    - But greedy never gave Arm 2 a chance
3. **Q-values converge to sample mean**
    - Q(1) started at 10.0 (two lucky pulls)
    - By play 20, Q(1) = 7.30 (closer to true mean 5.0)
    - But never had chance to explore other arms
4. **Regret grows indefinitely**
    - Each play on suboptimal arm = regret
    - Over 100 plays: regret would be ~100
    - Over 1000 plays: regret would be ~1000!

COMPARISON: GREEDY VS OPTIMAL

20 Plays

- **Greedy total:** 95
- **Optimal (all Arm 2):** 120
- **Regret:** 25

100 Plays

- **Greedy estimate:** ~475 (locked in Arm 1)
- **Optimal (all Arm 2):** 600
- **Regret:** 125

1000 Plays

- **Greedy estimate:** ~4750 (stuck on Arm 1, mean 5.0)
- **Optimal (all Arm 2):** 6000
- **Regret:** 1250

Greedy's weakness grows over time!

MATHEMATICAL FORMULA USED

Q-Value Update

$$Q_{new}(a) = \frac{Q_{old}(a) \times N_{old}(a) + R_t}{N_{new}(a)}$$

Where: - $Q_{new}(a)$ = updated estimate - $Q_{old}(a)$ = previous average - $N_{old}(a)$ = times pulled before - R_t = reward received - $N_{new}(a) = N_{old}(a) + 1$

Greedy Action Selection

$$a^* = \arg\max_a Q(a)$$

Select arm with highest estimated Q value.

SUMMARY

What Happened

1. Greedy algorithm started with all Q=0
2. Tried Arm 1, got lucky (reward 10)
3. Q(1) = 10.0 became best estimate
4. Locked in and exploited Arm 1 forever
5. Never discovered better Arm 2
6. Final reward: 95 (should have been 120+)

Why It Failed

- **No exploration:** Pure greedy never tries untested arms
- **Local optimum:** Gets stuck on first apparently-good arm
- **High regret:** Performance gap grows over time

Lesson for Next Approaches

All subsequent approaches (-greedy, optimistic, UCB) will add **exploration mechanisms** to avoid this trap.

CONCLUSION

Greedy approach is purely exploitative. While it achieves decent short-term reward (95 points in 20 plays), it completely fails to discover the optimal arm and suffers increasing regret over time.

The fundamental problem: With no exploration, greedy is helpless against early bad luck or early good luck on suboptimal arms. This motivates the need for exploration strategies in subsequent approaches.